Design of a Vending Machine using Finite Automata

Subject: Theory of Computing

Course: B.Tech CSE

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Abstract

This report presents the design of a vending machine using Finite Automata. It models a real-world system where coins are inserted and products are dispensed. The system is represented using DFA, showing how theoretical computation concepts apply practically.

Introduction

Finite Automata are used to model systems with finite states. A vending machine operates on inputs like coins and produces outputs like products. This project demonstrates applying TOC concepts to real systems.

Problem Statement

Design a vending machine that accepts ■5 and ■10 coins and dispenses a product at ■15. Assume no change is returned and only valid inputs are provided.

Theoretical Background

A DFA is defined as $(Q, \Sigma, \delta, q0, F)$. It consists of states, input symbols, transition function, start state and final states.

Methodology

States represent money collected. Transitions occur on coin inputs. Final state reached at ■15 triggers output.

Analysis

The model correctly accepts valid combinations (5+10, 10+5). It is deterministic and simple.

Applications

Used in ATMs, ticket machines, vending systems, and embedded systems.

Conclusion

The project shows how DFA can model real-world systems effectively.

Future Scope

Extend for multiple products, change return, and advanced automata.

References

Hopcroft TOC Book, GeeksforGeeks, NPTEL Lectures

State Transition Table

Current State	Input	Next State
q0	5	q1
q0	10	q2
q1	5	q2
q1	10	q3
q2	5	q3